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Unit-3
Oscillator and OPAMP

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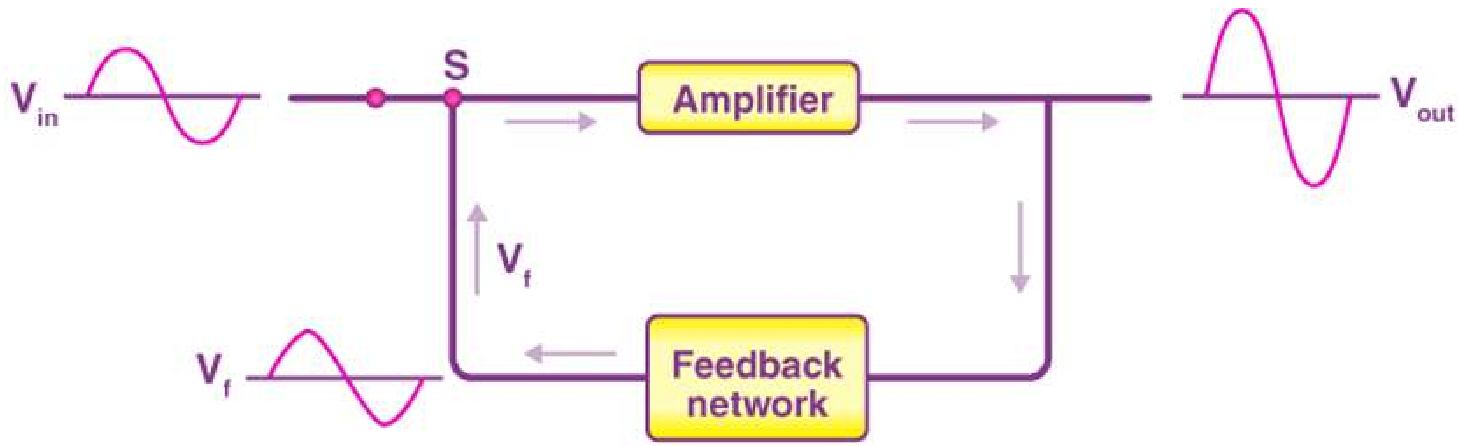
Feedback Amplifier

- The purpose of an amplifier is to amplify the input signal without changing its characteristics except its amplitude.

- The amplifier that works on the principle of feedback is called feedback amplifier .

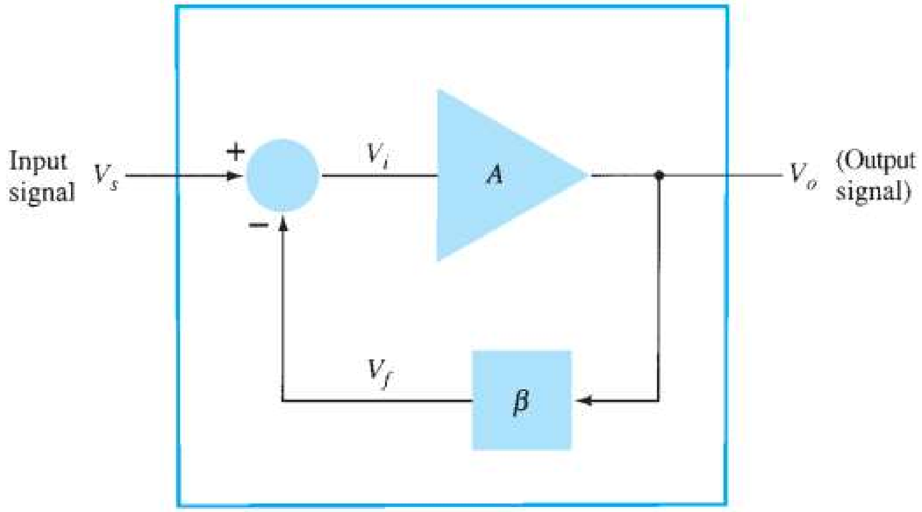
- Feedback is a process where a fraction of the output (voltage/current) is fed back to the input. A simple block diagram of a feedback amplifier is shown in Fig. 1



Feedback Amplifier

- The input signal Vs is applied to a mixer network, in which it is combined with a feedback signal Vf. The difference of these signals V*i* is the input to the amplifier.
- A portion of the amplifier output is connected to the input through a feedback network.
- Positive feedback is a process in which a portion of the output signal is fed back to the input in the same phase, thereby increasing the net input signal.
- Negative feedback is a process in which a portion of the output signal is fed back to the input in opposite phase, thereby reducing the net input signal.

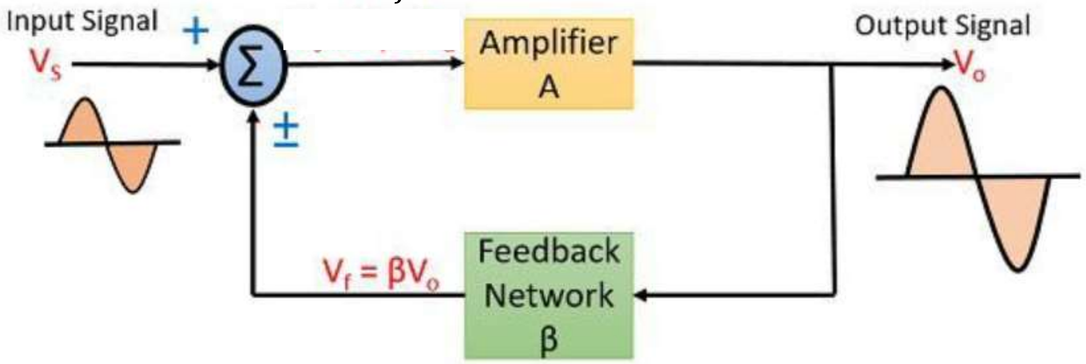
- A = Open-loop gain
- β = Feedback factor
- V_s = Source voltage
- V_i = Input to amplifier
- V_o = Output voltage
- V_f = Feedback voltage



Feedback amplifier

Positive and Negative Feedback

$$V_s \pm V_f = V_i$$



- A = Open-loop gain
- β = Feedback factor
- V_s = Source voltage
- V_i = Input to amplifier
- V_o = Output voltage
- V_f = Feedback voltage

For **negative feedback**:

$$V_i = V_s - \beta V_o$$

$$V_o = AV_i$$

$$V_o = A(V_s - \beta V_o)$$

$$V_o = AV_s - A\beta V_o$$

$$V_o(1 + A\beta) = AV_s$$

$$A_f = \frac{V_o}{V_s} = \frac{A}{1 + A\beta}$$

Closed-loop Gain: $A_f = \dfrac{V_o}{V_s} = \dfrac{A}{1 + A\beta}$

For **Positive feedback**:

$$V_i = V_s + \beta V_o$$

$$V_o = AV_i$$

$$V_o = A(V_s + \beta V_o)$$

$$V_o = AV_s + A\beta V_o$$

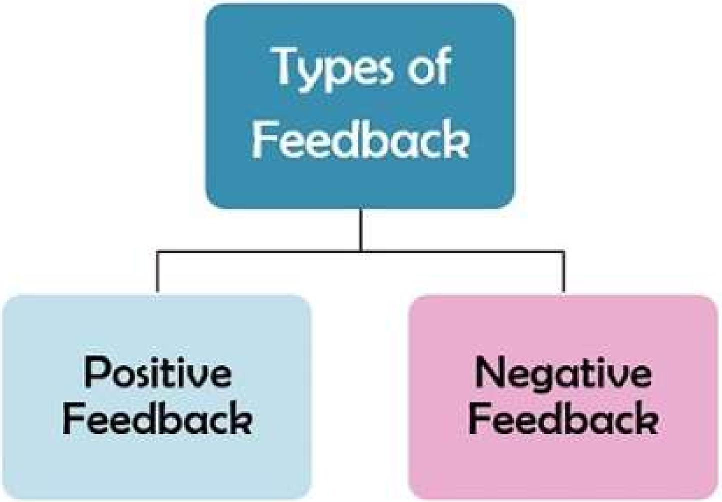
$$V_o - A\beta V_o = AV_s$$

$$V_o(1 - A\beta) = AV_s$$

$$A_f = \frac{V_o}{V_s}$$

$$\boxed{A_f = \frac{A}{1 - A\beta}}$$

Positive and Negative Feedback



Parameter	Negative Feedback	Positive Feedback
Gain Formula	$A_f = \frac{V_o}{V_s} = \frac{A}{1 + A\beta}$	$A_f = \frac{A}{1 - A\beta}$
Stability	Increases	Decreases
Distortion	Reduces	Increases
Used In	Amplifiers	Oscillators

What is Positive Feedback

•There is the in-phase relationship between input and output.
•It increases the net gain of the system.
•There is the use of oscillators.

What is Negative Feedback

•There is out-of-phase link between input and output for this system
•Its gain value is less.
•Its stability is higher than positive feedback.

Barkhausen Criterion for Oscillation Design

- The use of positive feedback that results in a feedback amplifier having closed-loop gain $|A\beta|$ greater than 1 and satisfies the phase conditions will result in operation as an oscillator circuit.
- An oscillator circuit is capable of producing ac voltage of desired frequency and waveshape.
- If the output signal varies sinusoidally, the circuit is referred to as a sinusoidal oscillator.
- If the output voltage rises quickly to one voltage level and later drops quickly to another voltage level, the circuit is generally referred to as a pulse or square-wave oscillator.

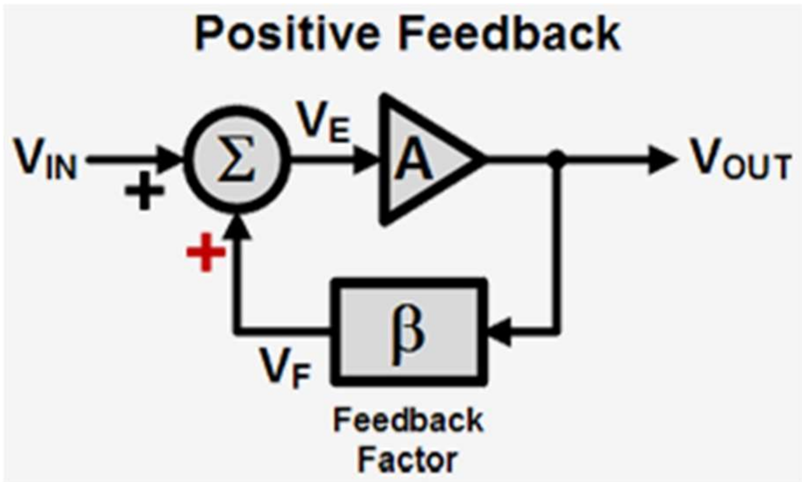
Barkhausen Criterion

1. Magnitude Condition

$$|A\beta| = 1$$

2. Phase Condition

$$\angle A\beta = 0^\circ \text{ or } 360^\circ$$

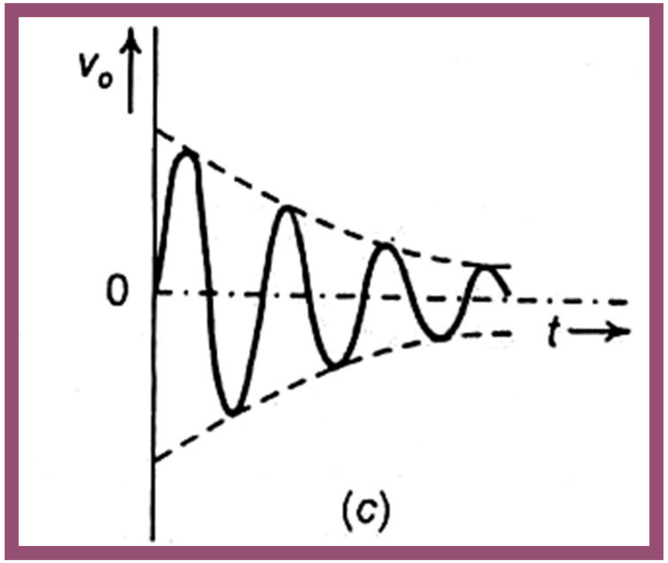


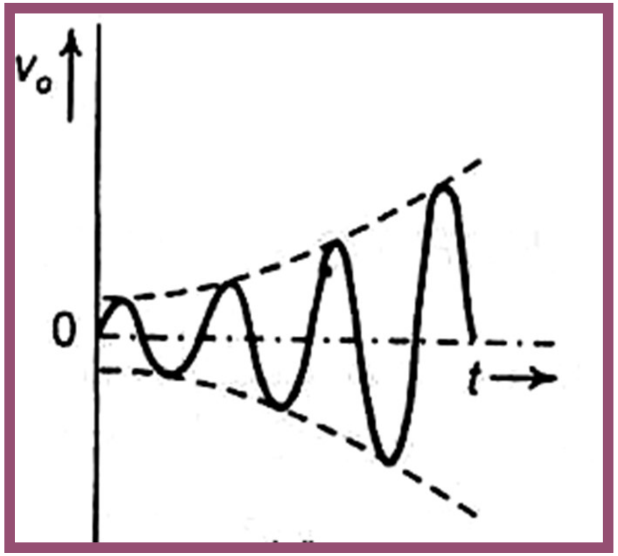
Barkhausen criterion for oscillation

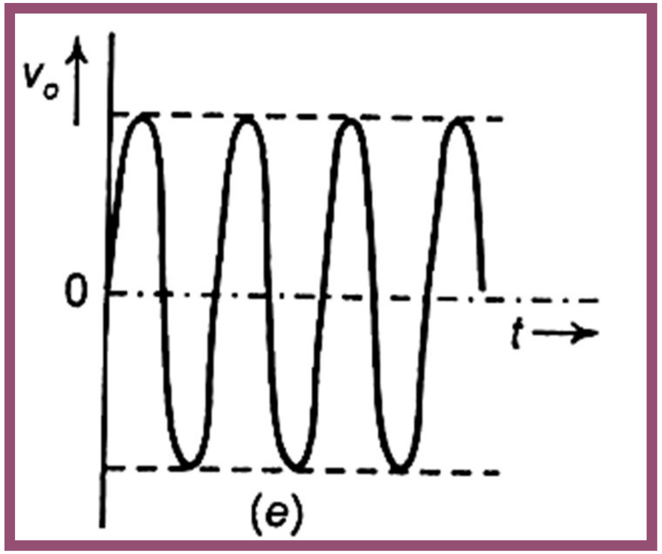
- The output waveform will still exist after the switch is closed if the condition is met $\beta A=1$
- This is known as the Barkhausen criterion for oscillation.
- In reality, no input signal is needed to start the oscillator going. Only the condition $\beta A=1$ must be satisfied for self-sustained oscillations to result.
- In practice, $\beta A > 1$system is started oscillating by amplifying noise voltage. The resulting waveforms are never exactly sinusoidal.
- The closer the value βA is to exactly 1, the more nearly sinusoidal is the waveform.

OSCILLATOR OPERATION

- $|A\beta| < 1$ → Oscillations die out
- $|A\beta| > 1$ → Amplitude grows (distortion)
- $|A\beta| = 1$ → Sustained oscillations



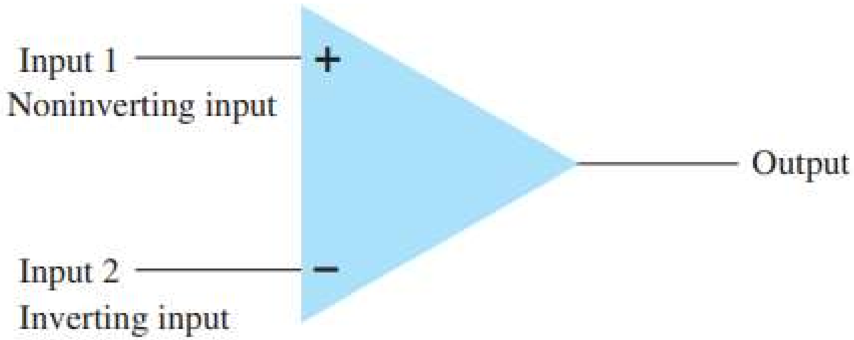




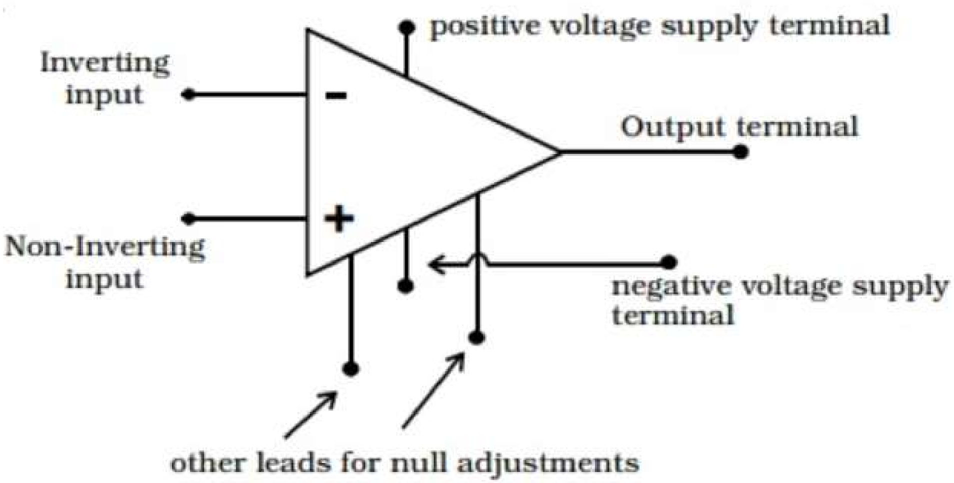
OPERATIONAL AMPLIFIERS

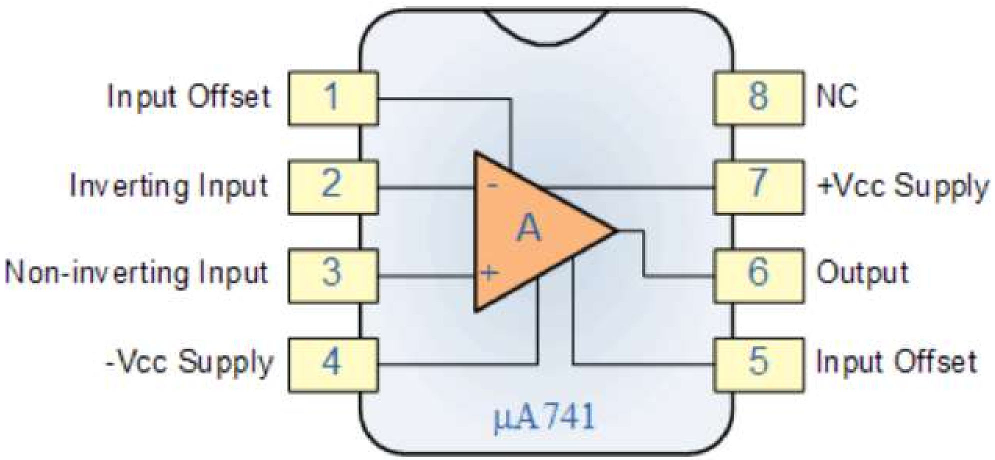
- An OP-AMP is so named, because it was originally designed to perform mathematical operations such as addition, subtraction, multiplication, division, integration, differentiation etc in analog computer.
- **Operational Amplifier (Op-Amp) :** An **Operational Amplifier (Op-Amp)** is a high-gain voltage amplifier with **two inputs** and **one output**. It amplifies the **difference between two input voltages**.

- The OP - AMP is represented by a triangular symbol as shown in Fig. It has two input terminals and one output terminal.
- The terminal with *negative* sign is called as the inverting input and the terminal with *positive* sign is called as the non-inverting input.



Basics of OPAMP





The output voltage is directly proportional to the difference voltage $V_{d.}$

$$V_{out} = A(V_+ - V_-)$$

Where:

- A = open-loop gain
- V_+ = non-inverting input voltage
- V_- = inverting input voltage

•Infinite open-loop gain ($A \to \infty$)
•Infinite input impedance (no input current)
•Zero output impedance
•Infinite bandwidth
•Zero offset voltage

Equivalent Circuit of OPAMP

- The circuit which represents op-amp parameters in terms of physical components, for the analysis purpose is called equivalent circuit of an op-amp.
- The circuit shows the op-amp parameters like input resistance, output resistance, the open loop voltage gain in terms of circuit components like R_{in}, R_o etc.
- The op-amp amplifies the difference between the two input voltages.

$V_o = A_{OL}V_d = A_{OL}(V_1 - V_2)$

where A_{OL} = Large signal open loop voltage gain.

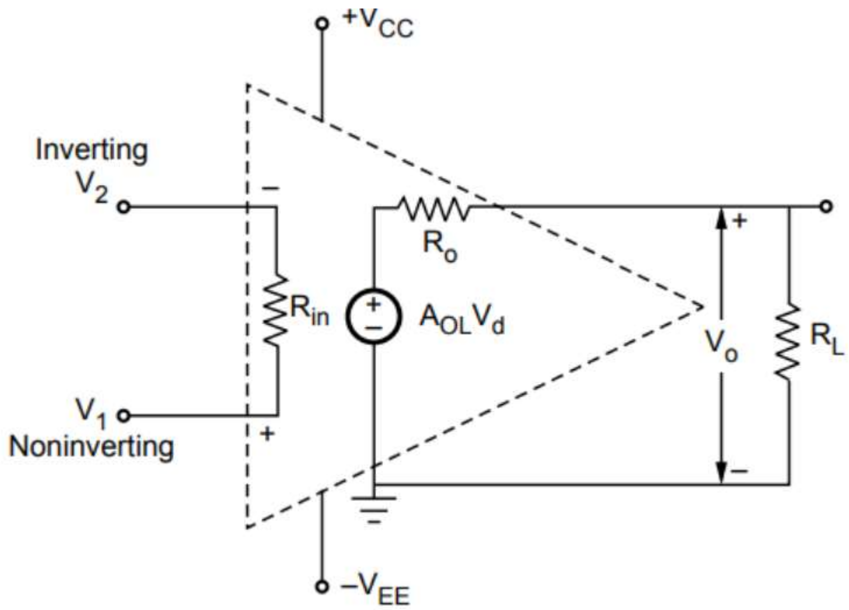
V_d = Difference voltage $V_1 - V_2$.

V_1 = Noninverting input voltage with respect to ground.

V_2 = Inverting input voltage with respect to ground.

R_i = Input resistance of op-amp.

R_o = Output resistance of op-amp.

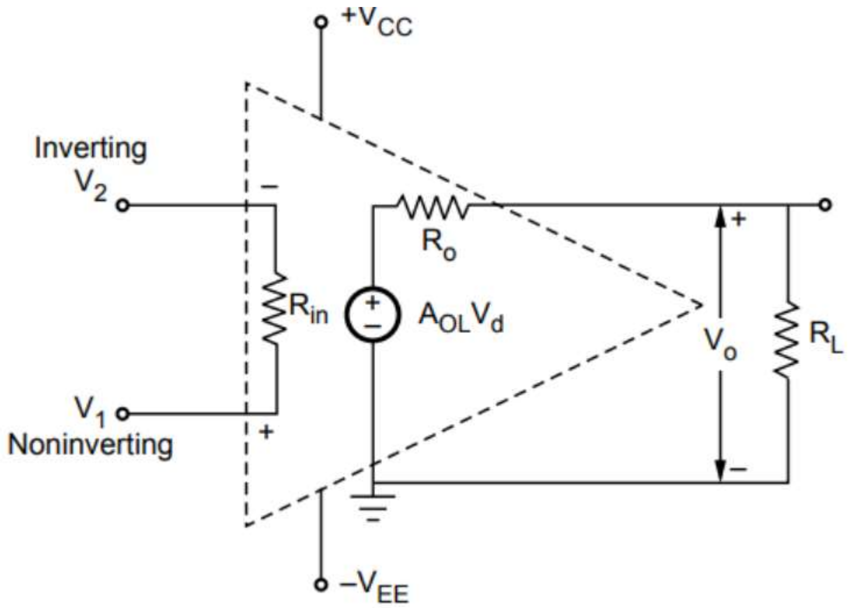


Equivalent Circuit of OPAMP

It is to be noted that the op-amp amplifies difference voltage and not the individual input voltages. Thus, the output polarity gets decided by the polarity of the difference voltage V_d.

$$V_o = A_{OL} V_d = A_{OL}(V_1 - V_2)$$

where A_{OL} = Large signal open loop voltage gain.



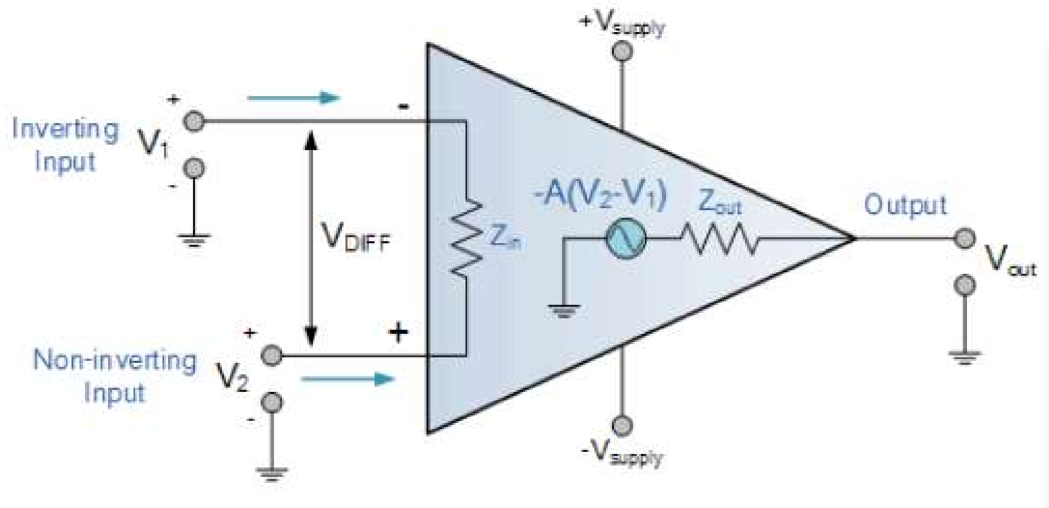
- The voltage source $A_{OL}V_d$ is the Thevenin's equivalent voltage source while Ro is the Thevenin's equivalent resistance looking back into the output terminals.

Characteristics of Ideal OPAMP

a) Infinite voltage gain : (A = ∞)
It is denoted as A. It Is the differential open loop gain and is infinite for an ideal op-amp.

b) Infinite input impedance : (Z_{in} = ∞)
The input impedance is denoted as Z_{in} and is infinite for an ideal op-amp. This ensures that no current can flow into an ideal op-amp.
An ideal op-amp draws no current at both the input terminals i.e. $I_1 = I_2 = 0$. Thus, its input impedance is infinite

c) Zero output impedance : (Z_{out} = 0)
The output impedance is denoted as Z_{out} and is zero for an ideal op-amp. This ensures that the output voltage of the op-amp remains same, irrespective of the value of the load resistance connected.



Characteristics of Ideal OPAMP

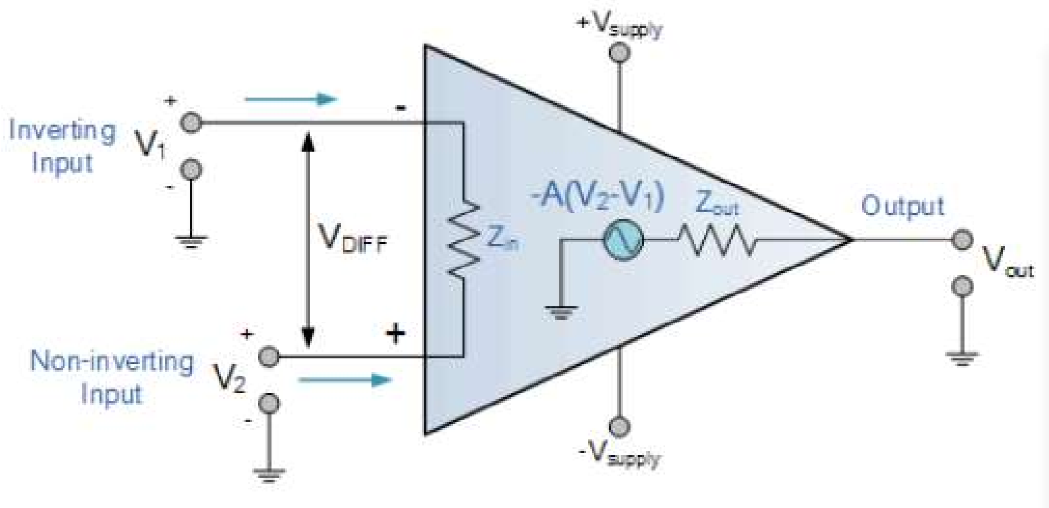
d) Zero offset voltage : (V_{ios} = 0)
The presence of the small output voltage though $V_1 = V_2 = 0$ is called an offset voltage. It is zero for an ideal op-amp. This ensures zero output for zero input signal voltage.

e) Infinite bandwidth : The range of frequency over which the amplifier performance is satisfactory is called its bandwidth. The bandwidth of an ideal op-amp is infinite.

f) Infinite slew rate : (S = ∞)
This ensures that the changes in the output voltage occur simultaneously with the changes in the input voltage.

Slew Rate: The Slew rate (SR) of an op-amp is defined as the maximum rate of change of output voltage per unit of time. It is represented as volts per microsecond ($V/\mu s$).

$$\boxed{SR = \frac{\Delta V_o}{\Delta t}\ \mathrm{V}/\mu\mathrm{s}} \quad \text{with } t \text{ in } \mu\mathrm{s}$$

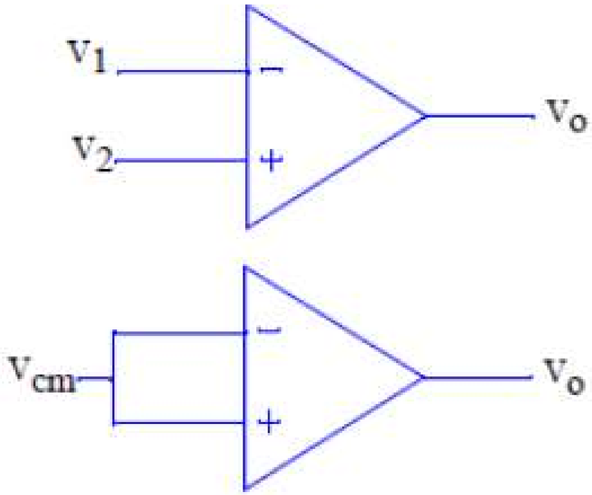


Characteristics of Ideal OPAMP

g) Infinite CMRR : (CMRR = ∞)

The common-mode rejection ratio (CMRR) in a differential amplifier serves as a measure of the amplifier's ability to reject unwanted common-mode signals, such as noise. It is defined as the ratio of the differential gain (Ad) to the common-mode gain (Acm).

- ❖ **Common-mode gain** A_{cm} is the gain of an amplifier when the **same signal is applied to both inputs**.
- ❖ **Differential gain** A_d is the gain of an amplifier when it amplifies the **difference between two input voltages**.



$$A_d = \frac{v_o}{v_2 - v_1}$$

$$A_{cm} = \frac{v_o}{v_{cm}}$$

$$CMRR_{dB} = 20\log_{10}\left(\frac{A_d}{A_{cm}}\right)$$

$$CMRR = \frac{A_d}{A_{cm}}$$

Comparison of Ideal and Practical OPAMP

Property	Ideal	Practical(Typical)
Open-loop gain	Infinite	Very high (>10000)
Open-loop bandwidth	Infinite	Dominant pole(≅10 Hz)
CMRR	Infinite	High (> 60 dB)
Input Resistance	Infinite	High (>1 MΩ)
Output Resistance	Zero	Low(< 100 Ω)
Input Bias Currents	Zero	Low (< 50 nA)
Offset Voltages	Zero	Low (< 10 mV)
Offset Currents	Zero	Low (< 50 nA)
Slew Rate	Infinite	A few V/µs

Applications of Op-Amps

Op-amps are used in a wide variety of applications, including:

- **Signal Amplification**: Used to increase the amplitude of weak signals.
- **Filters**: Used in active filters (low-pass, high-pass, band-pass, etc.) for audio and communication applications.
- **Mathematical Operations**: Used in circuits that perform integration, differentiation, addition, and subtraction.
- **Voltage Follower (Buffer)**: Provides high input impedance and low output impedance to isolate stages without affecting the signal.
- **Oscillators and Signal Generators**: Used to generate waveforms like sine, square, and triangular waves.
- **Analog Computation**: In analog computers, op-amps are used to perform arithmetic and calculus functions.

Inverting OPAMP

An **Inverting Operational Amplifier** (inverting op-amp) is basically fixed-gain voltage amplifier circuit configuration that amplifies an input signal while “inverting” its output signal by 180 degrees compared to its input.

In the inverting amplifier:

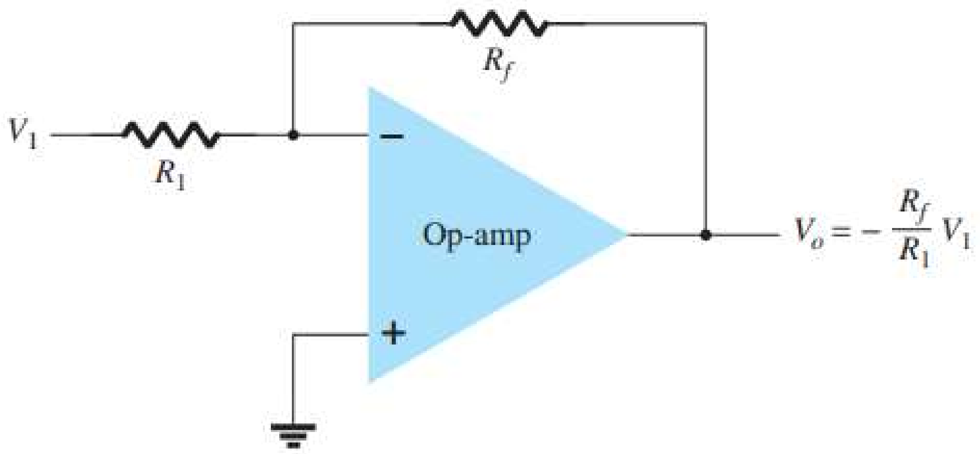
•The **non-inverting (+) terminal is grounded**.

•Due to **very high gain (A → ∞)** and **negative feedback**, the op-amp forces:

$$V_+ \approx V_-$$

Since $V_+ = 0V$,

$$V_- \approx 0V$$



The inverting terminal is **not physically connected to ground.** It behaves *like* ground (0V), but it is not actually connected to ground. This is called **Virtual Ground**.

Inverting OPAMP

- An input signal V_1 is applied through resistor R_1 to the minus input. The output is connected back to the same minus input through resistor R_f. The plus input is connected to ground.
- Since the signal V_1 is essentially applied to the minus input, the resulting output is opposite in phase to the input signal.
- The concept of a virtual short implies that although the voltage is nearly 0 V, there is no current through the amplifier input to ground.

No current enters op-amp input

$$I_1 = I_f$$

Current through input resistor

$$I_1 = \frac{V_1 - 0}{R_1}$$

Current through feedback resistor

$$I_f = \frac{0 - V_o}{R_f}$$

Apply KCL at Inverting Node

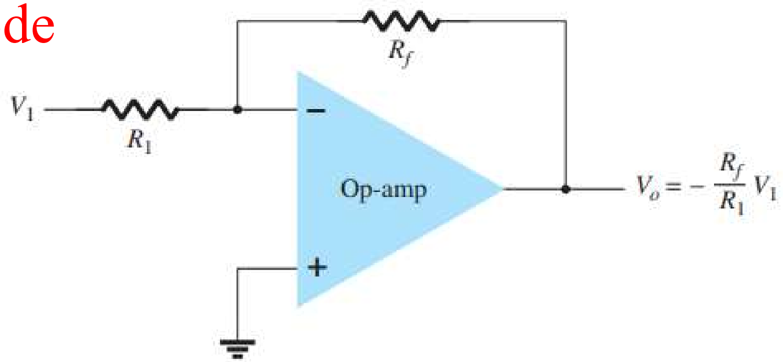
$$I_1 = I_f$$

$$\frac{V_1}{R_1} = -\frac{V_o}{R_f}$$

$$V_o = -\frac{R_f}{R_1} V_1$$

Gain of Inverting Amplifier

$$A_v = \frac{V_o}{V_i} = -\frac{R_f}{R_1}$$



- Output is 180° phase shifted
- Inverting node is at virtual ground

Inverting Amplifier: Characteristics and Applications

An inverting amplifier is an op-amp configuration where the input is applied to the inverting terminal, and the non-inverting terminal is grounded.

Characteristics

1. **Phase Inversion**: The output signal is 180 degrees out of phase with the input, meaning it is inverted.
2. **Gain**: The gain of an inverting amplifier is determined by the ratio of the feedback resistor

   $Av = -Rf/R1$

1. **High Input Impedance** (with feedback): Feedback helps achieve higher input impedance than the input resistor alone.
2. **Virtual Ground**: The inverting input is at a virtual ground, where it is at the same potential as ground (0V), but no current flows through it.

Applications

1. **Summing Amplifier**: An inverting amplifier can add multiple input signals, each with its own resistor.
2. **Analog Integration and Differentiation**: Used in analog signal processing for integration and differentiation of signals.
3. **Active Filters**: Inverting amplifiers are often used in active filter designs due to their controllable gain.
4. **Oscillators**: Inverting amplifiers can be used in oscillators to produce square wave or other periodic signals.

Non- Inverting OPAMP

The input signal V_1 is applied to the plus terminal, and the output is fed back to the minus terminal through resistor R_f with resistor R_1 connected to ground.
Due to the **virtual short**, the voltages at the plus and minus terminals are nearly equal, no current flows into the op-amp input.

Input is applied to non-inverting terminal: $V_+ = V_- = V_1$

Current through input resistor

$$I_1 = \frac{0 - V_1}{R_1} = -\frac{V_1}{R_1}$$

Current through feedback resistor

$$I_f = \frac{V_1 - V_o}{R_f}$$

Current through R_1 equals current through R_f.

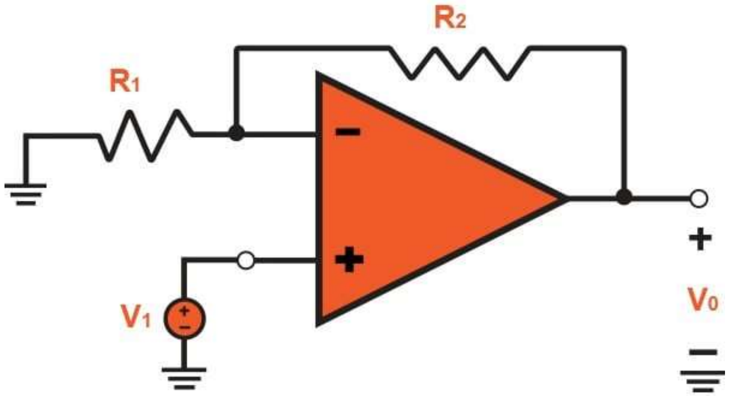
$$I_1 = I_f$$

$$-\frac{V_1}{R_1} = \frac{V_1 - V_o}{R_f}$$

$$\frac{R_f}{R_1} V_i = V_o - V_i$$

$$V_o = V_i + \frac{R_f}{R_1} V_i$$

$$V_o = \left(1 + \frac{R_f}{R_1}\right) V_i$$



$$A_v = \frac{V_o}{V_i} = 1 + \frac{R_f}{R_1}$$

Output is in phase with input

Non-Inverting Amplifier: Characteristics and Applications

A non-inverting amplifier is an op-amp configuration where the input is applied to the non-inverting terminal, and the output is fed back to the inverting terminal through a resistor.

Characteristics

- **No Phase Inversion**: The output signal is in phase with the input, so there is no inversion.
- **Gain**: The gain of a non-inverting amplifier is positive and given by: $A_{CL} = 1 + R_f/R_1$
- **High Input Impedance**: Non-inverting amplifiers provide extremely high input impedance, often in the megohms or higher range.
- **Excellent Signal Fidelity**: Since the output is in phase with the input, it provides accurate signal amplification without inversion.

Applications

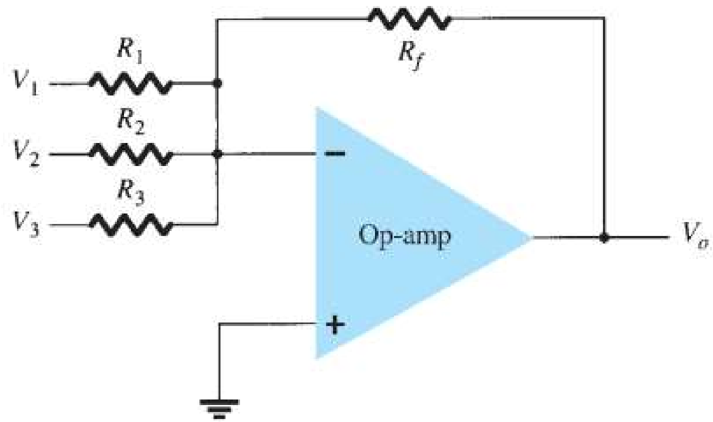
1. **Voltage Follower (Buffer)**: When Rf=0 and R1 is infinite, the gain becomes 1, and the circuit acts as a buffer, offering high input and low output impedance without altering the signal.
2. **Precision Amplifiers**: Non-inverting amplifiers are often used in applications requiring precise, non-inverted amplification, such as sensor signal amplification.
3. **Audio Preamplifiers**: Used in audio equipment to amplify signals without altering their phase, ensuring accurate sound reproduction.

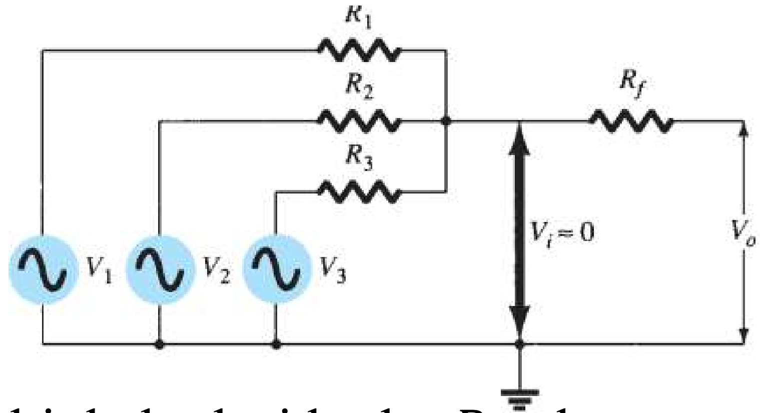
Data Acquisition: Amplifies weak signals from sensors in data acquisition systems without phase distortion, which is essential for accurate measurements.

Feature	Inverting Amplifier	Non-Inverting Amplifier
Input Terminal	Inverting (-)	Non-inverting (+)
Output Phase	180° phase shift (inverted)	In phase with input
Gain	$-R_f/R_1$	$1+R_f/R_1$
Input Impedance	Moderate	High
Applications	Summing, integration, filters, oscillators	Voltage follower, precision amplification

Summing Amplifier using OPAMP

The circuit shows a three-input summing amplifier circuit, which provides a means of algebraically summing (adding) three voltages, each multiplied by a constant-gain





$$I_1 + I_2 + I_3 = I_f$$

$$\frac{V_1 - 0}{R_1} + \frac{V_2 - 0}{R_2} + \frac{V_3 - 0}{R_3} = \frac{0 - V_o}{R_f}$$

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = \frac{-V_o}{R_f}$$

Multiply both sides by R_f: the output voltage can be obtained as

$$R_f\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}\right) = -V_o$$

$$\boxed{V_o = -R_f\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}\right)}$$

$$\boxed{V_o = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3\right)}$$

PHASE-SHIFT OSCILLATOR

A **Phase Shift Oscillator** is a sinusoidal oscillator that produces a **stable sine wave** using an **amplifier (op-amp or transistor)** and an **RC phase-shift network**.
The RC network provides the required **phase shift** so that the feedback becomes **positive feedback**, which is necessary for oscillation.

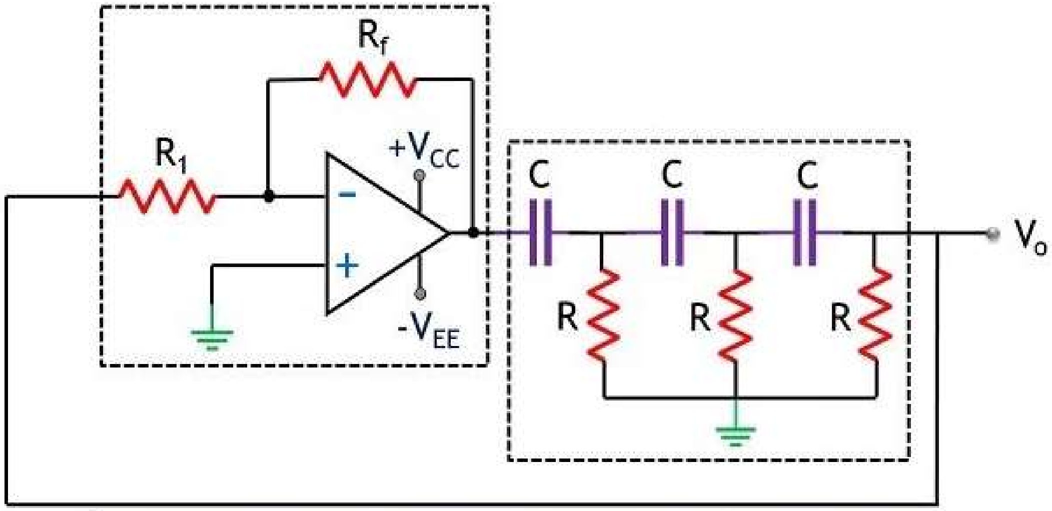
For oscillation must βA> 1 and phase shift around the feedback network is 180°(providing positive feedback).

For the loop gain βA to be greater than unity, the gain of the amplifier stage must be greater than 1/ β or 29

$$\boxed{f = \frac{1}{2\pi RC\sqrt{6}}}$$

$$\beta = \frac{1}{29}$$

$$A > 29$$



Circuit of Phase Shift Oscillator using Op-amp

The values of R and C to provide (at a specific frequency) 60°-phase shift per section for three sections, resulting in a 180°phase shift, as desired.

PHASE-SHIFT OSCILLATOR

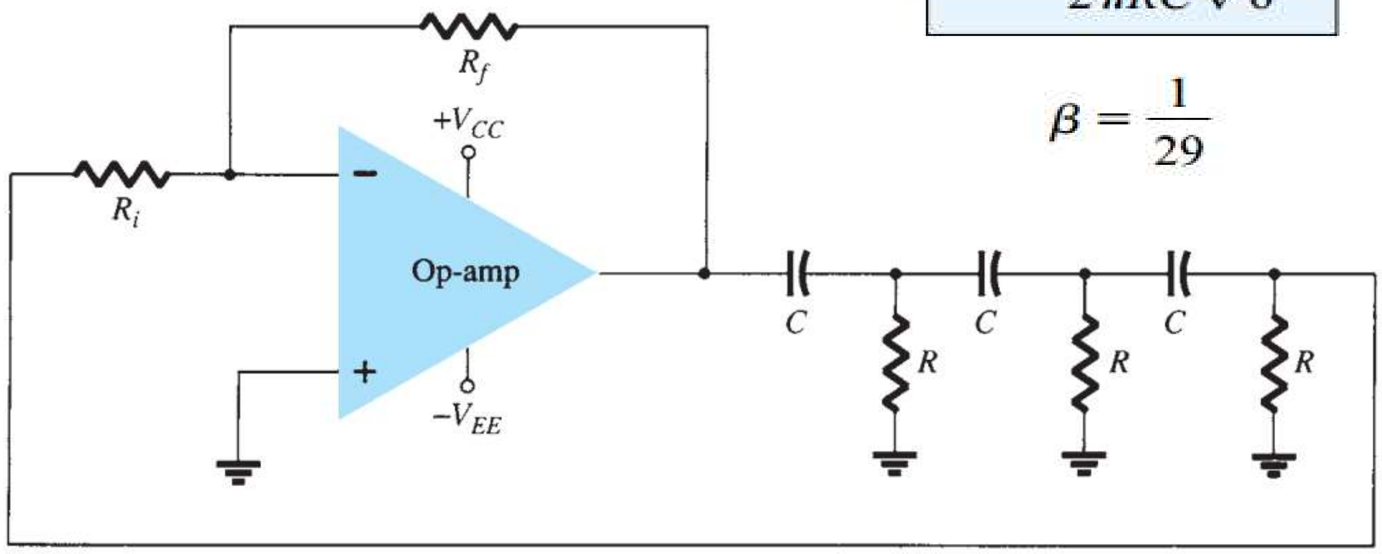
- The output of the op-amp is fed to a three-stage RC network, which provides the needed 180° of phase shift (at an attenuation factor of 1/29).
- If the op-amp provides gain (set by resistors Ri and Rf) of greater than 29, a loop gain greater than unity results and the circuit acts as an oscillator

$$\boxed{f = \frac{1}{2\pi RC\sqrt{6}}}$$

βA> 1

$$\beta = \frac{1}{29}$$

$$A > 29$$



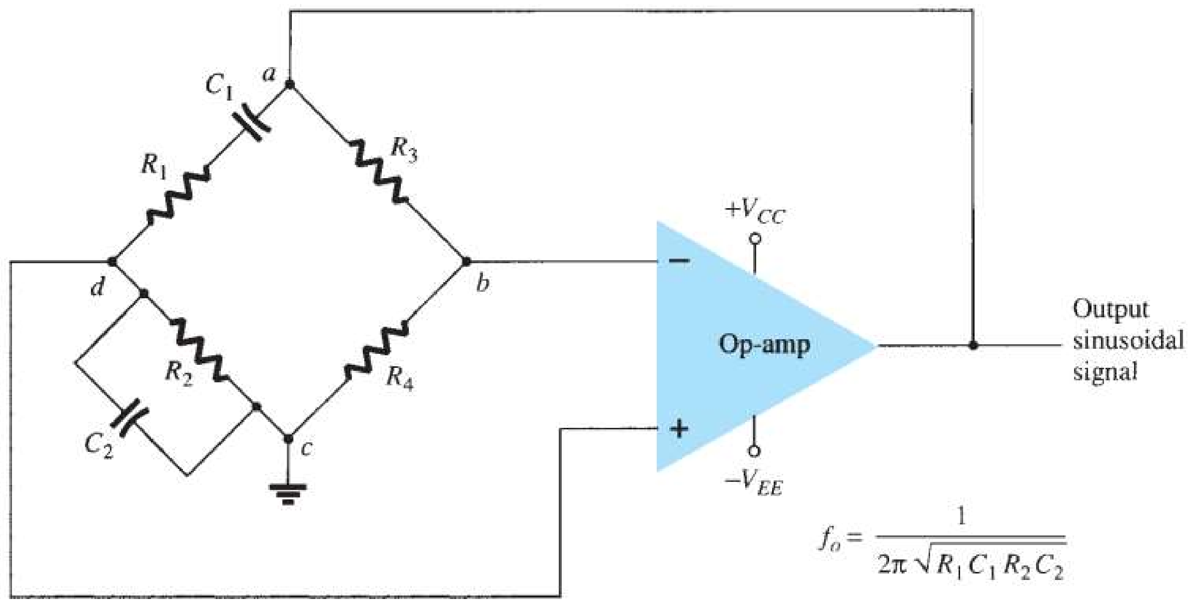
WIEN BRIDGE OSCILLATOR

- A **Wien Bridge Oscillator** is a **sinusoidal oscillator** that uses an **RC bridge network and an operational amplifier (op-amp)** to generate **low-distortion sine waves**.
- It is widely used in **audio frequency generators (20 Hz – 1 MHz)**.

➢Resistors R1 and R2 and capacitors C1 and C2 form the frequency-adjustment elements, while resistors R3 and R4 form part of the feedback path.

➢The op-amp output is connected as the bridge input at points a and c.

➢The bridge circuit output at points b and d is the input to the op-amp.



$$\frac{Z_1}{Z_2} = \frac{Z_3}{Z_4}$$

In a **Wien Bridge Oscillator**, the **resonant frequency** (or oscillation frequency) is the frequency at which the **Wien bridge RC network becomes balanced and produces zero phase shift** between the input and output signals.

WIEN BRIDGE OSCILLATOR

At resonant frequency frequency:

- Phase shift = **0°**
- Feedback fraction = **1/3**

$$\beta = \frac{1}{3}$$

Using **Barkhausen criterion**

$$A\beta = 1$$

$$A \times \frac{1}{3} = 1$$

$$A = 3$$

Therefore required amplifier gain:

$$A = 1 + \frac{R_3}{R_4} = 3$$

$$\therefore R_3 = 2R_4$$

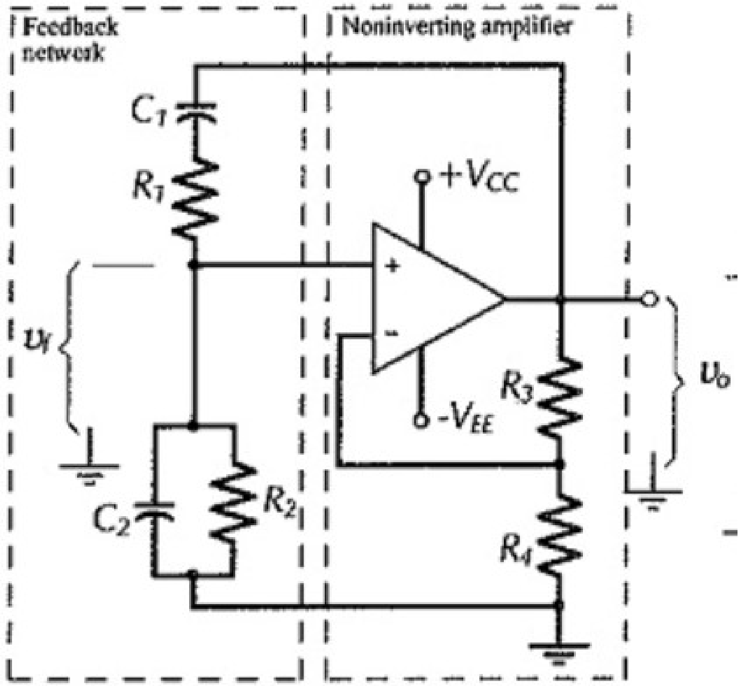
$$\frac{R_3}{R_4} = \frac{R_1}{R_2} + \frac{C_2}{C_1}$$

$$\boxed{f_o = \frac{1}{2\pi\sqrt{R_1C_1R_2C_2}}}$$

If, in particular, the values are $R_1 = R_2 = R$ and $C_1 = C_2 = C$, the resulting oscillator frequency is

$$\boxed{f_o = \frac{1}{2\pi RC}}$$

$$\boxed{\frac{R_3}{R_4} = 2}$$



Thus, a ratio of R3 to R4 greater than 2 will provide sufficient loop gain for the circuit to oscillate at the frequency f_0.

Advantages and Applications

Advantages

1.The overall gain of the oscillator is high as it uses a two-stage amplifier.

2.As no inductors are used in the circuit, there is no issue of interference from external magnetic fields.

3.This oscillator produces a stable sinewave without any distortions.

4.The frequency of the oscillations can be changed by changing the values of capacitors or by the use of a variable resistor in the circuit.

5.The Wein-bridge oscillator has good frequency stability.

Applications

- Audio signal generators
- Function generators
- Electronic test equipment
- Communication systems
- Instrumentation circuits

Example 12.4 The RC network of a Wien bridge oscillator consists of resistors and capacitors of values $R_1 = R_2 = 22\ \text{k}\Omega$ and $C_1 = C_2 = 100\ \text{pF}$. Calculate the frequency of oscillaton.

Solution :

Here, $R = 22\ \text{k}\Omega = 22 \times 10^3\ \Omega$ and $C = 100\ \text{pF} = 10^{-10}\ \text{F}$.

$$f_0 = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 22 \times 10^3 \times 10^{-10}} = \mathbf{72.34\ \ kHz}$$